

Homework 5

Due Time: 2:00 pm, June 1 (Monday)

Problem 1. Suppose that a mathematical model fits linear programming except for the following restrictions.

1. At least one of the following two inequalities holds:

$$4x_1 - x_2 - x_3 + x_4 \leq 14,$$

$$x_1 + 2x_2 + x_3 + x_4 \leq 18.$$

2. At least two of the following three inequalities hold:

$$2x_1 + 4x_2 - x_3 + x_4 \leq 32,$$

$$-x_1 + 3x_2 + 4x_3 + x_4 \leq 42,$$

$$3x_1 - x_2 + 3x_3 - 2x_4 \leq 58.$$

Show how to reformulate these restrictions to fit an MIP model.

Problem 2. Consider the following integer nonlinear programming problem:

$$\begin{array}{ll} \text{Maximize} & Z = 5x_1^2 - x_1^3 + 9x_2^2 - x_2^4, \\ \text{subject to} & x_1 + x_2 \leq 3, \\ & x_1, x_2 \geq 0, \\ & x_1, x_2 \text{ are integers.} \end{array}$$

This problem can be reformulated as an equivalent pure BIP problem with a linear objective function using six binary variables y_{ij} , for $i = 1, 2$ and $j = 1, 2, 3$.

1. Formulate a BIP model for this problem where the binary variables have the interpretation

$$y_{ij} = \begin{cases} 1, & \text{if } x_i = j, \\ 0, & \text{otherwise.} \end{cases}$$

2. Formulate a BIP model for this problem where the binary variables have the interpretation

$$y_{ij} = \begin{cases} 1, & \text{if } x_i \geq j, \\ 0, & \text{otherwise.} \end{cases}$$

Problem 3. Fly-Right Airplane Company has recently received purchase requests from three customers. Because of limited production capacity, the company cannot accept all requests. The relevant data are given below.

	Customer		
	1	2	3
Start-up cost	\$4 million	\$2 million	0
Marginal net revenue	\$2.2 million	\$2.8 million	\$0.9 million
Capacity used per plane	25%	35%	20%
Maximum order	3 planes	2 planes	5 planes

Formulate a mixed-integer programming model to determine how many airplanes to produce for each customer in order to maximize total profit.

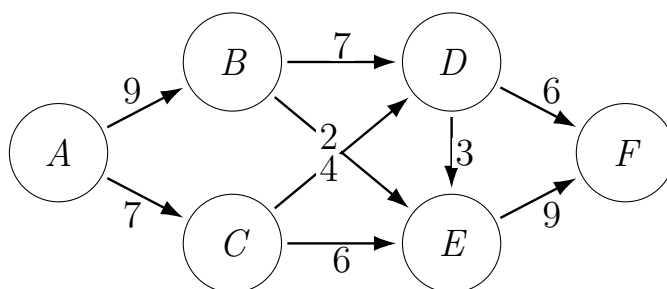
Problem 4. Four cargo ships will be used for shipping goods from one port to four other ports, labeled 1, 2, 3, 4. Any ship can be used for making any one of these four trips. The total costs are given in the following table:

Ship	Port			
	1	2	3	4
1	500	400	600	700
2	600	600	700	500
3	700	500	700	600
4	500	400	600	600

The objective is to assign the four ships to four different ports so as to minimize the total cost.

1. Formulate this problem as an integer programming model.
2. Solve this problem by considering its LP relaxation.
3. Explain why Simplex method still gives an integer optimal solution for this type of problem even after relaxing the binary restrictions.

Problem 5. Consider the maximum flow problem shown below. The source is node A , the sink is node F , and the arc capacities are the numbers shown next to the directed arcs.



1. Use the augmenting path method to solve this maximum flow problem. List each augmenting path, the amount of flow augmented along that path, and the final flow on each arc.
2. Formulate this maximum flow problem as a linear programming problem. Define the decision variables, objective function, capacity constraints, and flow conservation constraints explicitly.
3. Prove that the solution obtained in part (1) is optimal.